

SHYMKENT, Kazakhstan

WMO: 383280

Lat: 42.3478N

Lon: 69.7100E

Elev: 1982

StdP: 13.67

Time Zone: 6.00 (E06)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	10.4	-0.9	5.6	5.5	4.8	7.6	13.1	18.9	41.2	16.1	38.1	3.6	120	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.6	100.7	65.9	98.4	65.4	95.4	64.6	69.1	91.8	67.7	90.5	66.5	89.7	7.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	90.1	77.2	60.3	84.0	75.9	58.6	79.0	75.0	34.5	92.1	33.3	89.9	32.3	89.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	15.2	13.3	0.1	105.9	8.2	2.9	-5.8	108.0	-10.6	109.7	-15.2	111.3	-21.2	113.4
			-0.5	71.6	7.9	2.3	-6.2	73.3	-10.8	74.6	-15.3	75.9	-21.0	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	31.9	35.7	48.2	58.2	68.2	76.6	80.5	79.3	69.1	56.3	44.3	33.8
	DBStd	19.11	10.32	11.09	9.31	7.86	6.32	5.68	5.11	5.35	6.95	8.50	9.51	9.90
	HDD50	1901	565	410	143	22	1	0	0	0	0	39	216	505
	HDD65	4556	1027	822	522	227	40	2	0	1	37	286	624	968
	CDD50	4435	2	8	87	267	565	798	945	908	572	236	44	3
	CDD65	1616	0	0	2	22	139	351	480	443	159	18	2	0
	CDH74	21779	0	0	46	302	1665	4576	6715	6042	2110	300	23	0
	CDH80	11506	0	0	6	64	626	2422	3907	3463	938	76	4	0
Wind	WSAvg	5.6	4.7	5.4	6.0	6.4	6.4	6.3	5.9	6.3	5.8	4.9	4.6	4.7
Precipitation	PrecAvg	23.6	3.0	3.3	3.2	2.8	2.2	0.9	0.5	0.2	0.4	1.6	2.8	2.8
	PrecMax	32.5	6.1	7.6	5.7	6.2	5.6	2.7	3.2	0.9	2.2	3.7	6.4	6.1
	PrecMin	15.5	0.9	0.5	1.5	0.3	0.1	0.0	0.0	0.0	0.0	0.3	0.6	0.6
	PrecStd	4.4	1.3	1.5	1.1	1.3	1.3	0.6	0.6	0.2	0.5	1.2	1.9	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	67.6	79.2	85.9	93.5	101.8	105.9	104.1	96.8	87.4	77.3	62.6
		MCWB	48.1	51.8	56.0	61.9	64.4	66.6	66.2	65.6	63.9	59.8	57.1	48.3
	2%	DB	53.7	59.6	73.3	81.0	89.8	98.4	102.0	100.3	93.1	81.0	68.2	56.7
		MCWB	44.6	48.3	54.9	60.9	63.6	65.5	66.5	65.5	62.6	57.3	52.7	46.2
	5%	DB	49.1	54.9	68.1	77.3	86.5	95.2	98.6	97.2	89.6	76.9	62.7	51.7
		MCWB	42.1	45.8	52.9	58.7	62.7	65.0	65.7	64.8	61.0	54.6	50.3	44.1
	10%	DB	44.9	50.1	63.4	73.2	83.3	91.8	95.3	95.0	85.8	72.2	58.6	46.7
		MCWB	40.0	43.2	51.2	57.3	62.2	64.1	65.0	64.2	59.3	53.6	48.2	41.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	52.8	58.7	65.8	69.0	70.8	71.8	70.7	66.5	60.9	58.2	50.1
		MCDB	57.4	63.7	70.7	79.6	85.7	91.6	93.9	95.2	92.3	81.9	71.7	58.5
	2%	WB	45.6	49.9	56.4	62.9	66.5	68.4	69.1	68.3	63.9	58.5	54.7	47.4
		MCDB	52.3	57.9	69.6	76.1	83.7	89.8	93.0	92.6	89.3	77.0	65.0	55.2
	5%	WB	43.0	46.8	54.2	60.7	64.8	66.9	67.6	66.9	61.9	56.7	51.8	44.4
		MCDB	47.7	53.0	65.7	73.7	81.5	88.2	91.8	92.0	85.8	72.8	61.2	50.3
	10%	WB	40.5	43.9	52.2	58.7	63.2	65.6	66.3	65.4	60.1	54.8	49.1	41.5
		MCDB	44.6	49.4	62.2	70.2	79.5	86.9	91.0	91.1	83.1	69.2	56.6	46.4
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.4	20.9	21.7	24.6	27.2	27.6	28.5	28.2	24.4	19.2	17.3
		MCDBR	21.6	23.7	28.2	28.6	29.8	31.0	31.3	31.6	32.6	31.0	25.6	22.8
		MCWBR	14.3	14.5	14.1	14.4	12.9	11.9	11.2	12.0	13.0	13.5	13.5	14.5
	5% WB	MCDBR	19.3	21.4	24.6	24.9	26.8	28.0	28.3	29.7	31.2	26.7	22.1	20.7
		MCWBR	13.3	13.9	13.4	13.6	12.9	11.9	11.5	12.5	13.3	12.7	12.8	13.6
Clear-Sky Solar Irradiance	taub	0.348	0.387	0.446	0.460	0.438	0.424	0.409	0.381	0.368	0.416	0.402	0.344	
	taud	2.094	2.025	1.958	1.987	2.109	2.170	2.217	2.278	2.267	2.042	2.033	2.106	
	Ebn at Noon	239	249	248	256	266	270	273	278	272	235	214	227	
	Edh at Noon	34	43	52	54	49	47	44	40	37	42	36	31	
All-Sky Solar Radiation	RadAvg	563	843	1191	1643	2089	2366	2357	2122	1659	1050	626	458	
	RadStd	73	102	102	181	134	101	101	73	67	97	86	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+2.52	N/A	N/A	N/A	N/A	+329	+239
Regional (9 neighbors)	N/A	N/A	N/A	+1.04	N/A	-1.04	N/A	N/A	+194	+140

Nomenclature: See separate page